

5.79. Two samples of sizes 4 and 8 are drawn from a normally distributed population. Is the probability that one variance is greater than 1.5 times the other greater than 0.05, between 0.05 and 0.01, or less than 0.01?

variance of population = σ^2

sample variance S_1

sample variance S_2

$$a) F = \frac{mS_1^2/(m-1)\sigma^2}{nS_2^2/(n-1)\sigma^2}$$

$$= \frac{4S_1^2/3}{8S_2^2/7} = \frac{7}{6} \frac{S_1^2}{S_2^2}$$

$$S_1^2 > 1.5S_2^2$$

$$F = \frac{7}{6} \frac{1.5S_2^2}{S_2^2} = 1.75$$

$$F_{0.95} = 4.35$$

$$F_{0.99} = 8.45$$

$$b) F = \frac{8S_1^2/7}{4S_2^2/3} = \frac{6}{7} \frac{S_1^2}{S_2^2} = \frac{6}{7} 1.5 = 1.286$$

$$F_{0.95} = 8.89$$

$$F_{0.99} = 27.7$$

$\Rightarrow$ Both cases show that the ^{probability of} ratio of the two variances greater 1.5 are greater than 0.05.

Frequency distribution

5.82. Table 5-16 shows a frequency distribution of the lifetimes of 400 radio tubes tested at the L & M Tube Company. With reference to this table, determine the

- upper limit of the fifth class
- lower limit of the eighth class
- class mark of the seventh class

Table 5-16

Lifetime (hours)	Number of Tubes
300-399	14
400-499	46
500-599	58
600-699	76
700-799	68
800-899	62
900-999	48
1000-1099	22
1100-1199	6
TOTAL	400

- (d) class boundaries of the last class
- (e) class interval size
- (f) frequency of the fourth class
- (g) relative frequency of the sixth class
- (h) percentage of tubes whose lifetimes do not exceed 600 hours
- (i) percentage of tubes with lifetimes greater than or equal to 900 hours
- (j) percentage of tubes whose lifetimes are at least 500 but less than 1000 hours

- a) upper limit of fifth class = 799 hours
- b) lower limit of eighth class = 1000 hours
- c) class mark of the seventh class = $\frac{900+998}{2} = 949.5$ hours
- d) upper boundary of last class = $\frac{1199+1200}{2} = 1199.5$ hours
lower boundary of last class = $\frac{1099+1100}{2} = 1099.5$ hours
- e) class interval size = $1199.5 - 1099.5 = 100$ hours
- f) 76
- g) relative frequency of the sixth class = $\frac{62}{400} \times 100\% = 15.5\%$
- h) percentage of tubes < 600 hrs = $\frac{58+46+14}{400} \times 100\% = 29.5\%$
- i) percentage of tubes ≥ 900 hrs = $\frac{48+22+6}{400} \times 100\% = 19\%$
- j) $500 \leq$ percentage of tubes < 1000 hrs = $\frac{58+76+68+62+48}{400} \times 100\% = 78\%$

5.87. Table 5-17 shows the diameters in inches of a sample of 60 ball bearings manufactured by a company. Construct a frequency distribution of the diameters using appropriate class intervals.

Table 5-17

0.738	0.729	0.743	0.740	0.736	0.741	0.735	0.731	0.726	0.737
0.728	0.737	0.736	0.735	0.724	0.733	0.742	0.736	0.739	0.735
0.745	0.736	0.742	0.740	0.728	0.738	0.725	0.733	0.734	0.732
0.733	0.730	0.732	0.730	0.739	0.734	0.738	0.739	0.727	0.735
0.735	0.732	0.735	0.727	0.734	0.732	0.736	0.741	0.736	0.744
0.732	0.737	0.731	0.746	0.735	0.735	0.729	0.734	0.730	0.740

Lowest value = 0.724 Highest value = 0.746

We divide the data into 6 classes with class interval of 0.005

Diameter	Frequency
0.720 - 0.724	1
0.725 - 0.729	8
0.730 - 0.734	17
0.735 - 0.739	23
0.740 - 0.744	9
0.745 - 0.749	2
total	60

5.107. (a) Find the mean and standard deviation for the data of Problem 5.28.

(b) Construct a frequency distribution for the data, and find the standard deviation.

(c) Compare the result of (b) with that of (a).

$$a) \bar{x} = \frac{\sum x}{40} = \frac{5872}{40} = 146.8 \text{ lb}$$

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{40}} = \sqrt{\frac{66424}{40}} = 12.89 \text{ lb}$$

b) Lowest value = 119 Highest value = 176 range = 57 lb

Weight (lb)	Frequency	Class mark	$f(x - \bar{x})^2$
119 - 128	4	123.5	2025
129 - 138	7	133.5	1093.75
139 - 148	13	143.5	81.25
149 - 158	9	153.5	506.25
159 - 168	5	163.5	1531.25
169 - 178	2	173.5	1512.5
total	40	—	6750

$$\begin{aligned} \bar{x} &= \frac{123.5(4) + 133.5(7) + 143.5(13) + 153.5(9) + 163.5(5) + 173.5(2)}{40} \\ &= \frac{5840}{40} \\ &= 146 \text{ lb} \end{aligned}$$

$$\sigma^2 = \frac{\sum f(x - \bar{x})^2}{n} = \frac{6750}{40} = 12.99 \text{ lb}$$

- c) The value of mean in (a) is slightly larger than in (b), by 0.8 to be exact. The value of standard deviation in (b) is slightly larger than in (a), by 0.1. This error is due to the utilisation of class mark when calculating both mean and standard deviation. However, the margin error is small which indicates that (b) is a better method when dealing with large sample size as the calculation needed is lesser.